

Model Paper - Database Management Systems

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Q1 (20 marks)

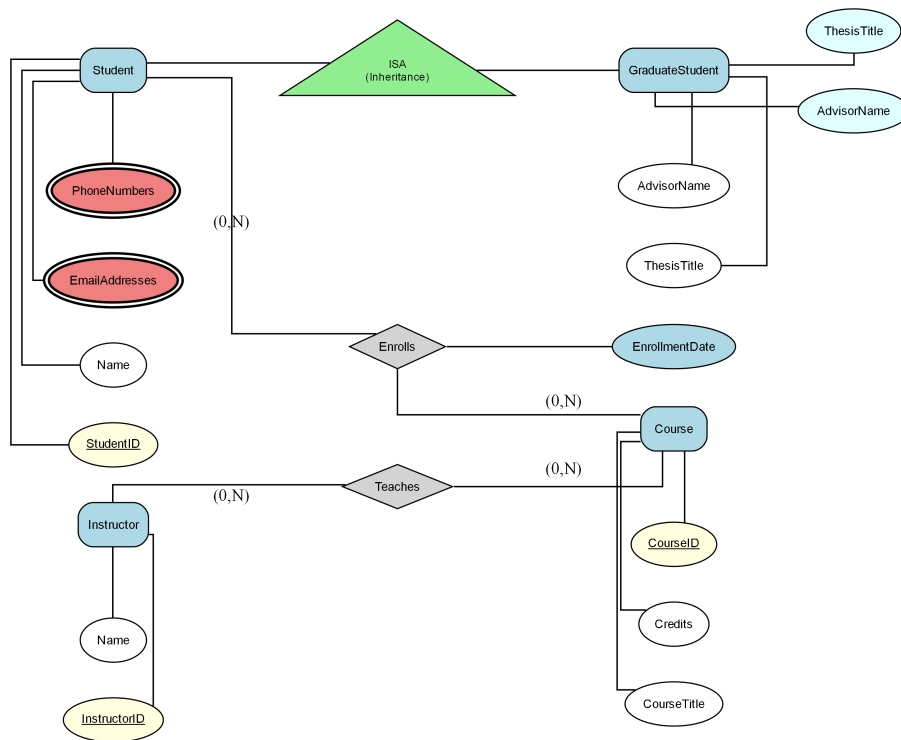
Consider the following Enhanced Entity-Relationship (EER) diagram:

Note: The diagram uses (min, max) cardinality notation where:

- (1,1) indicates one-to-one relationship (each entity participates exactly once)
- (1,N) or (1,*) indicates one-to-many relationship (one entity can relate to many)
- (0,1) indicates optional participation (zero or one)
- (0,N) or (0,*) indicates optional many participation (zero or many)

A university has several entities involved in the academic community. The entities include Student, Course, and Instructor. Each Student has attributes such as StudentID, Name, EmailAddresses, and PhoneNumbers. Courses have CourseID, CourseTitle, and Credits. Instructors have InstructorID and Name. Each Student can enroll in multiple Courses, and each Course can be taught by multiple Instructors. The enrollment relationship includes an EnrollmentDate attribute. Additionally, there is a specialization where GraduateStudent is a subtype of Student, with the attributes ThesisTitle and AdvisorName. A Student can be related to many Courses, and a Course can be related to many Students through the 'Enrolls' relationship (Student participation is optional, Course participation is optional). A Instructor can be related to many Courses, and a Course can be related to many Instructors through the 'Teaches' relationship (Instructor participation is optional, Course participation is optional)

Figure: EER



a) Convert the following EER model into the relational model. Indicate the primary keys and the foreign keys of the resulting relations clearly. (17 marks)

b) What is the best option for mapping the ISA hierarchies in the diagram? Justify your answer. (3 marks)

Q2 (15 marks)

Consider a relation $R(A, B, C, D, E)$ with the following set of functional dependencies F over R :

$F = \{A \rightarrow B, B \rightarrow C, A \rightarrow CD, D \rightarrow E\}$. Analyze this relation to identify keys and evaluate its normalization status.

a) Find all the keys in relation R using the attribute closure method. (4 marks)

b) Is R in 3NF? Give reasons for your conclusion. (3 marks)

c) Is R in BCNF? Give reasons for your conclusion. If R is not in BCNF, convert it to a set of BCNF relations. (8 marks)

Q3 (25 marks)

A university is designing a database to manage its student enrollment system. The database should include tables for Students, Courses, and Enrollments. Each student has a unique student ID, name, and date of birth. Each course has a unique course ID, title, and description. Enrollments link students to the courses they are taking, recording the enrollment date. You are tasked with advancing the design and management strategies for this database system.

a) Write SQL queries to in type 2 driver, jdbc api calls are converted to native java api calls. accept or refute the above statement justifying your answer. (2 marks)

b) Code Segment: ````java String sql = "SELECT * FROM employees WHERE department = ?";`

PreparedStatement pstmt = connection.prepareStatement(sql); pstmt.setString(1, "IT"); ResultSet rs = pstmt.executeQuery(); `` Which type of JDBC statement is used in the code segment shown above? Briefly explain when this type of statement will be used. (2 marks)

c) Briefly explain how data integrity can be ensured in the student enrollment database design. (2 marks)

d) Briefly explain the options available for transferring data between two sources such as between tables and servers. (3 marks)

e) A university is establishing role-based access control for its database system that manages Courses, Students, and Enrollments. Sarah is the senior database administrator responsible for overall database administration. Emily is a junior database administrator responsible for managing the Enrollments database. Nathan is a database developer responsible for creating and maintaining database objects. Michael is a data entry operator responsible for entering and updating student enrollment data.

i. Write a T-SQL statement to create a login to Sarah with windows authentication. (2 marks)

ii. Provide Sarah with the necessary permissions to handle all administrative tasks within the server using a fixed server role. (3 marks)

iii. Assuming Emily's username is 'emily.e', write T-SQL statements to assign her the responsibility of managing the Enrollments database using a user-defined server role. (5 marks)

iv. Assuming Nathan's login name is 'nathan.n', write T-SQL statements to grant him permission to create tables and databases. (3 marks)

v. Assuming Michael's login name is 'michael.m', write T-SQL statements to allow him to perform data entry tasks on the Students and Enrollments databases. (3 marks)

Q4 (40 marks)

Consider the following schema of a database designed for a library: Book (bookId: int, title: varchar(100), author: varchar(50), isbn: varchar(20), publicationYear: int) Borrower (borrowerId: int, name: varchar(100), phone: varchar(15), email: varchar(50)) Loan (loanId: int, bookId: int, borrowerId: int, loanDate: date, returnDate: date) Fine (fineId: int, loanId: int, dueDate: date, amount: real, status: varchar(20)). The 'Book' table stores information about books available in the library. The 'Borrower' table stores information about library members. The 'Loan' table records details of books borrowed by members. The 'Fine' table tracks fines associated with overdue loans. The 'Loan' table manages book loans, with each loan having a unique ID and being associated with a specific book and member. It records the loan date, due date, and return date. The 'Fine' table contains information about fines incurred by members for late returns or other penalties. It includes a fine ID, member ID, fine amount, and payment status.

a) Write SQL Queries to perform the following:

i. Find the title of books who are linked to book records where title is 'The Great'. (4 marks)

ii. Find the book who has the highest total amount compared to all other books. Find the bookId and title. (6 marks)

iii. Find the title and title from Book, Book, and Loan where Loan.loanId is not null. (7 marks)

b) b) Create a function to calculate the total fine amount for a given book. This function should account for entity with a late penalty of 10% applied to those marked as 'overdue' payment status. (11 marks)

c) c) Create a trigger that automatically updates the 'amount' column in the 'Fine' table whenever a new fine record is added, or an existing fine record is updated. The 'amount' column should contain the total fine amount for each entity. Ensure the trigger adjusts the 'amount' column accurately to reflect any partial payments made by entity. The entity pays only 50% of the total fine amount as an initial payment when the payment status is marked as 'Partial'. (12 marks)